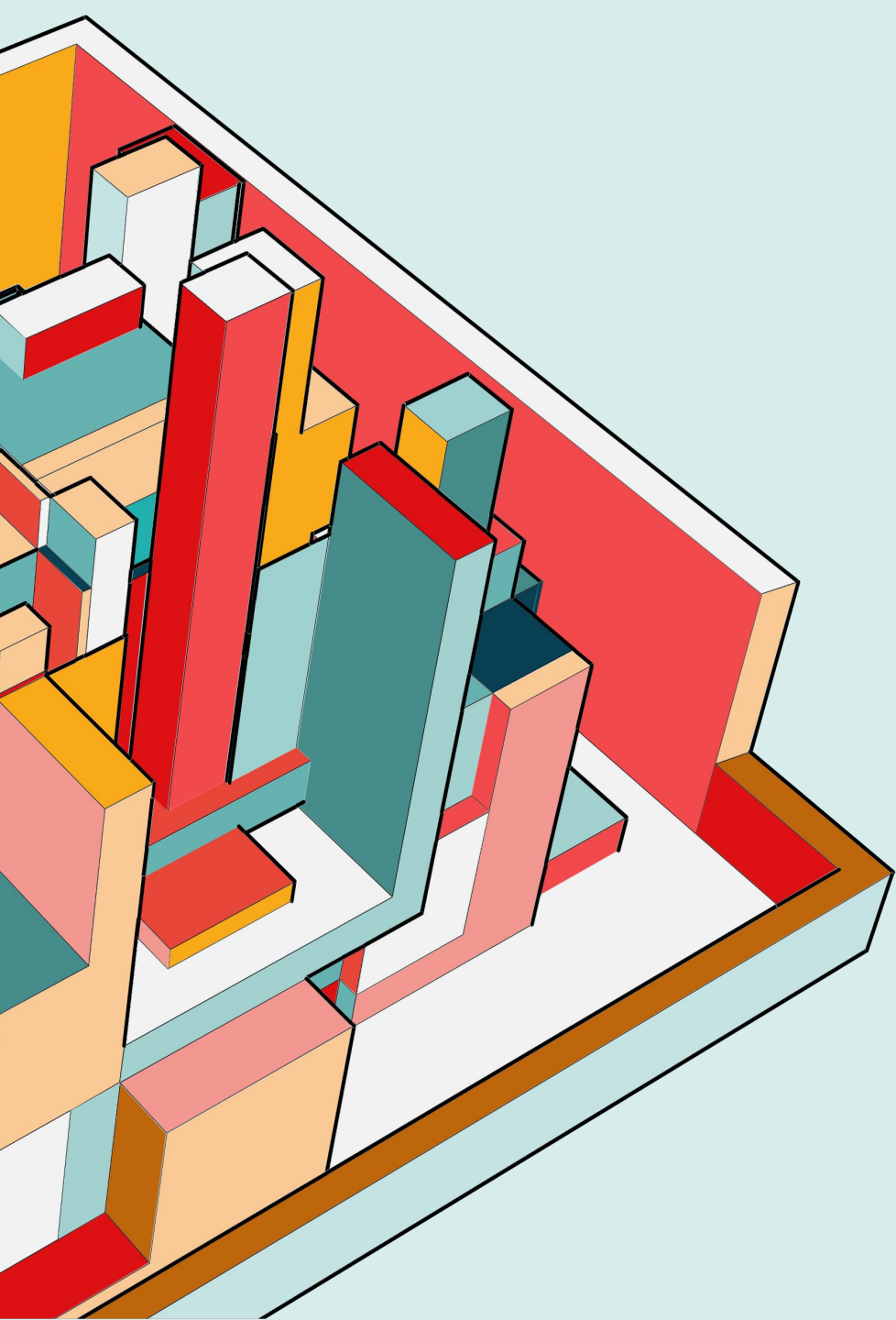




Methods 3 - Multilevel Statistical Modelling & Machine Learning

Classroom Instruction

WEEK 37

AGENDA

1. Welcome back from holidays :-)
2. Resources
3. Preamble + Setting up R and Python
5. Exercises for today

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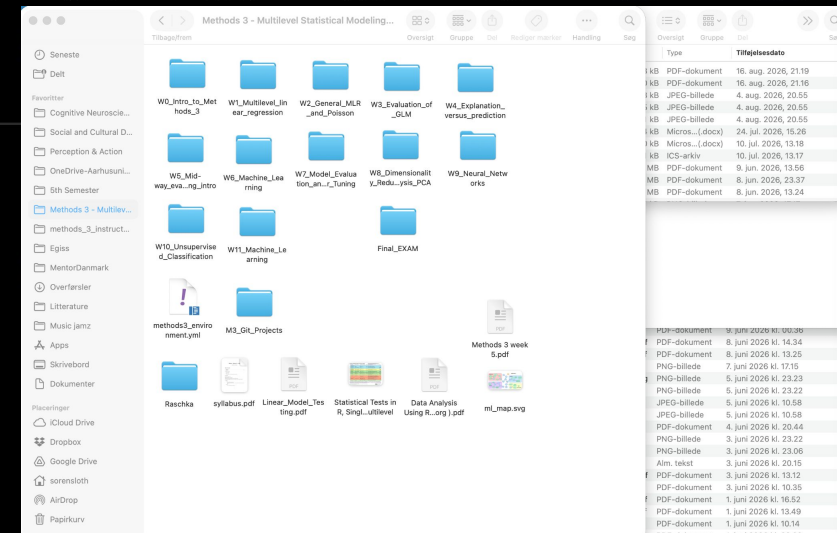
25

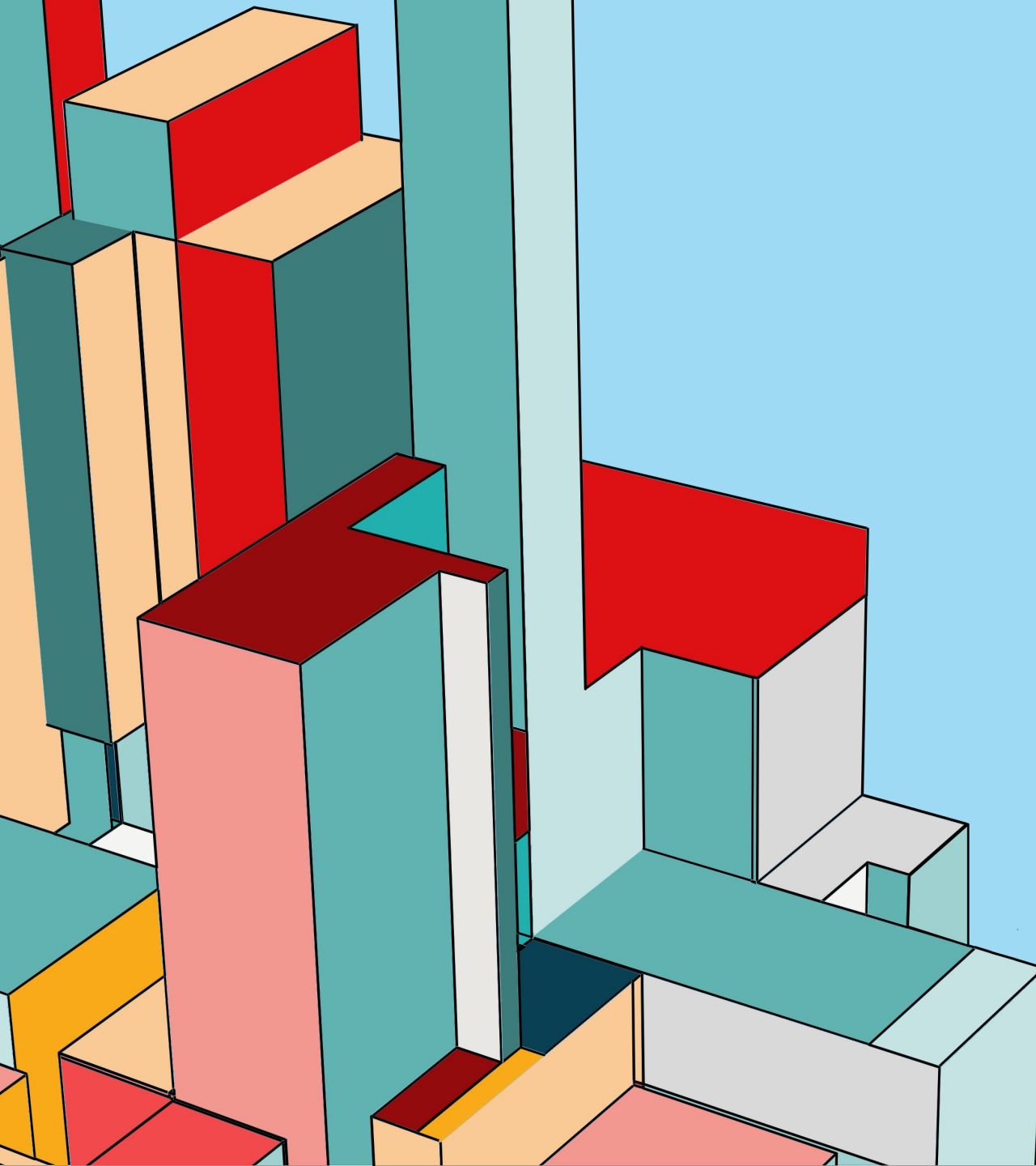
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RESOURCES

Resources

Go to GitHub classroom:

<https://github.com/Methods-3/Methods-3-F26/tree/main/assignments/preamble>

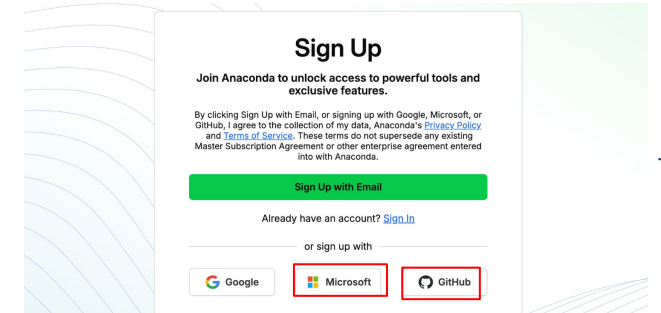
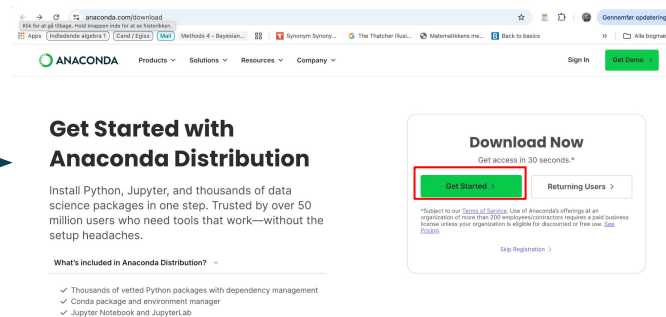
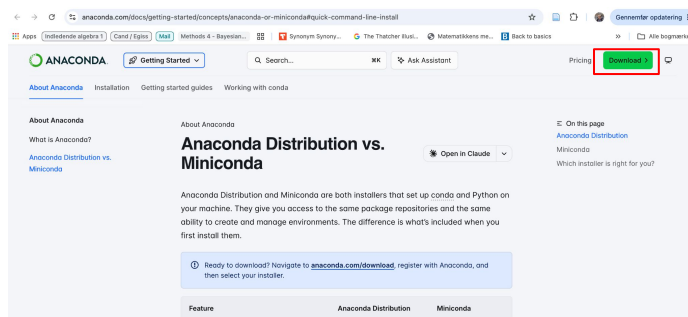
Download these 2 files:

- *methods3_environment.yml*
- *preamble.pdf*

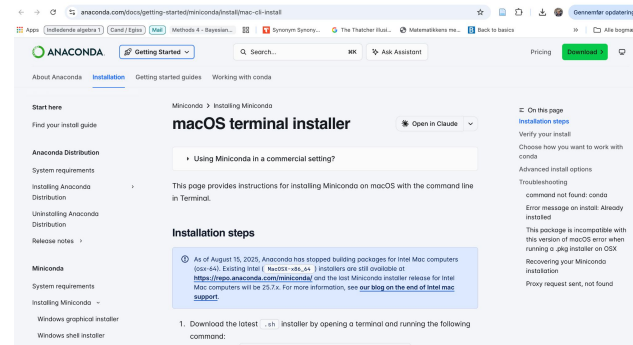
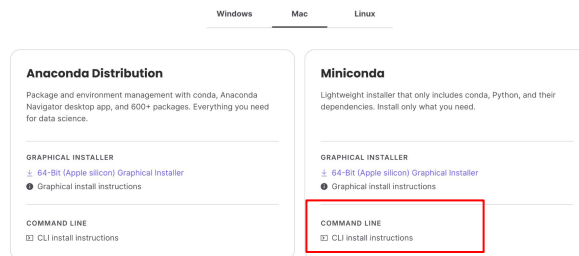
Do not download *preamble_test.pdf* !!!!

Follow the instructions in *preamble.pdf* first. Use me + slides for guidance.

Installing miniconda



Choose Your Download



```
(base) sorensloth@Mac methods_3_instructor_2026 % bash Miniconda3-latest-MacOSX-arm64.sh

Welcome to Miniconda3 py314_26.5.3-2

In order to continue the installation process, please review the license
agreement.
Please, press ENTER to continue
>>> MINICONDA END USER LICENSE AGREEMENT

Copyright Notice: Miniconda(R) (C) 2015, Anaconda, Inc.
All rights reserved. Miniconda(R) is licensed, not sold.

Redistribution and use in source and binary forms, with or without modification,
are permitted provided that the following conditions are met:

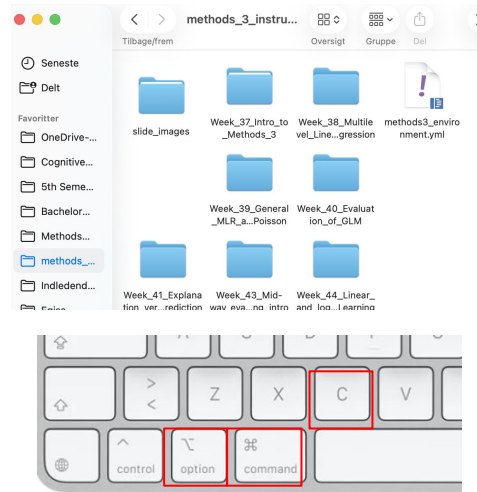
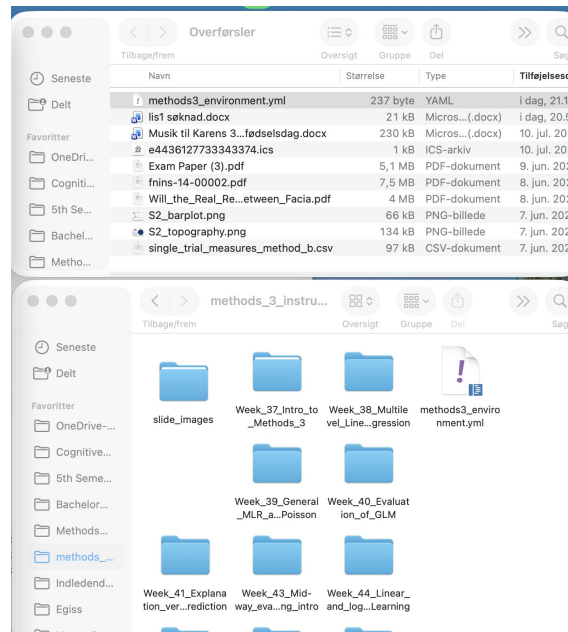
1. Redistributions of source code must retain the above copyright notice, this 1
```

Follow instructions...

*go to the directory with the installed
Miniconda3-file and ONLY write:
bash Miniconda3-latest-MacOSX-arm64.sh

if your terminal looks something like this, you
are on the right path. Follow through with the
instructions in the shell...

Installing miniconda



```
(base) sorensløth@Mac methods_3_instructor_2026 % cd /Users/sorensløth/Library/CloudStorage/OneDrive-Aarhusuniversitet/Cognitive Science Bach/5th Semester/methods_3_instructor_2026
```

```
conda env create -f methods3_environment.yml # create environment
conda activate methods3_2025 # activating your newly created environment
```

```
conda env create -f methods3_environment.yml
conda activate methods3_environment.yml
```

Create a "Methods_3" directory (folder) anywhere you'd like in your file manager. Pro tip: Actually name the directory something like Methods_3... Move the methods3_environment.yml to THAT folder.

1. Click on the folder background of the methods 3 directory
2. Press Option + Command + c
3. In terminal write: cd
4. Press Command + v
5. Enter to set directory

Run the lines one at a time in terminal.

After this follow the remaining steps in preamble.pdf

Cool extra thing

If you want your new "Methods_3" directory (or what you named it) to have a directory for each week and the portfolio assignments, then you can set your working directory

```
cd "PATH_TO_YOUR_METHODS_3_DIRECTORY"
```

Pro tip: you can copy the relative path with: option + command + c. The command `pwd` will print your working directory so you may check that you navigated correctly

And then paste in and ENTER

```
mkdir "Week_37_Intro_to_Methods_3" "Week_38_Multilevel_Linear_Regression" "Week_39_General_MLR_and_Poisson" "Week_40_Evaluation_of_GLM"
"Week_41_Explanation_versus_Prediction" "Week_43_Mid-way_evaluation_and_machine_learning_intro"
"Week_44_Linear_and_logistic_regression_Machine_Learning" "Week_45_Model_Evaluation_and_Hyperparameter_Tuning"
"Week_46_Dimensionality_Reduction_and_Principal_Component_Analysis_PCA" "Week_47_Neural_Networks" "Week_48_Unsupervised_Classification"
"Week_49_Final_Evaluation" "Week_50_Student_Presentations" "Portfolio_Assignments"
```

Pro tip: Change the name of the directories above before you paste and ENTER

Organizing Calendar

If you haven't already synchronized your calendar, maybe later today is the time. Synchronize through:

<https://timetable.au.dk/schedule>

OR

<https://mitstudie.au.dk/>

timebureau.au.dk/schedule

Apps [Cand / Engelsk](#) [Mail](#) [Methods 4 - Bayesian...](#) [Cognitive Neuroscience](#) [MUSIC](#) [Synonym Synonym...](#) [The Thatcher Illusion](#) [Matematikkens me...](#) [Back to basics](#) [Alle bogmærker](#)

timebureau.au.dk

Lokalebookingsportal | Mobil | Lokalevisning | Oversigtsvisning | Beskeder | English | Indstillinger | Hjælp | Log ud (780784)

Dag Uge Måned Liste

august 2026

Vis alle aktivitetstyper

Tilføj timeplan

AU Timebables

- 147102 Kognitionsvidenskab, bachelor 4. per. (1)
 - ☒ Afdelingsnede - LIC5 - ARTS-AHB F27
 - ☐ Cognitive Neuroscience F27 Aarhus - 147201U016
 - ☐ Cognitive Neuroscience F28 Aarhus - 147201U019
 - ☐ Metode 4: Bayesian Computationale Modellering F27 Aarhus - 147201U017
 - ☐ Metode 4: Bayesian Computationale Modellering F28 Aarhus - 147201U017
 - ☐ Social and Cultural Dynamics F27 Aarhus - 147201U018
 - ☐ Social and Cultural Dynamics F28 Aarhus - 147201U018
 - ☒ Undervisningsdag F27 - ARTS-AHB F27
 - ☐ Methods 1: Introduction to Experimental Methods, Statistics, and Programming E26 Aarhus - 147201U009
 - ☐ Methods 2: The General Linear Model F27 Aarhus - 147201U011
 - Søren Skott
 - ☒ Bachelor's project E26 Aarhus - 147201U015
 - ☒ Indledende algebra 1 E26 Aarhus - 85021U001

aug-2026						
M	T	W	T	F	S	S
27	28	29	30	31	1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31	1	2	3	4	5	6

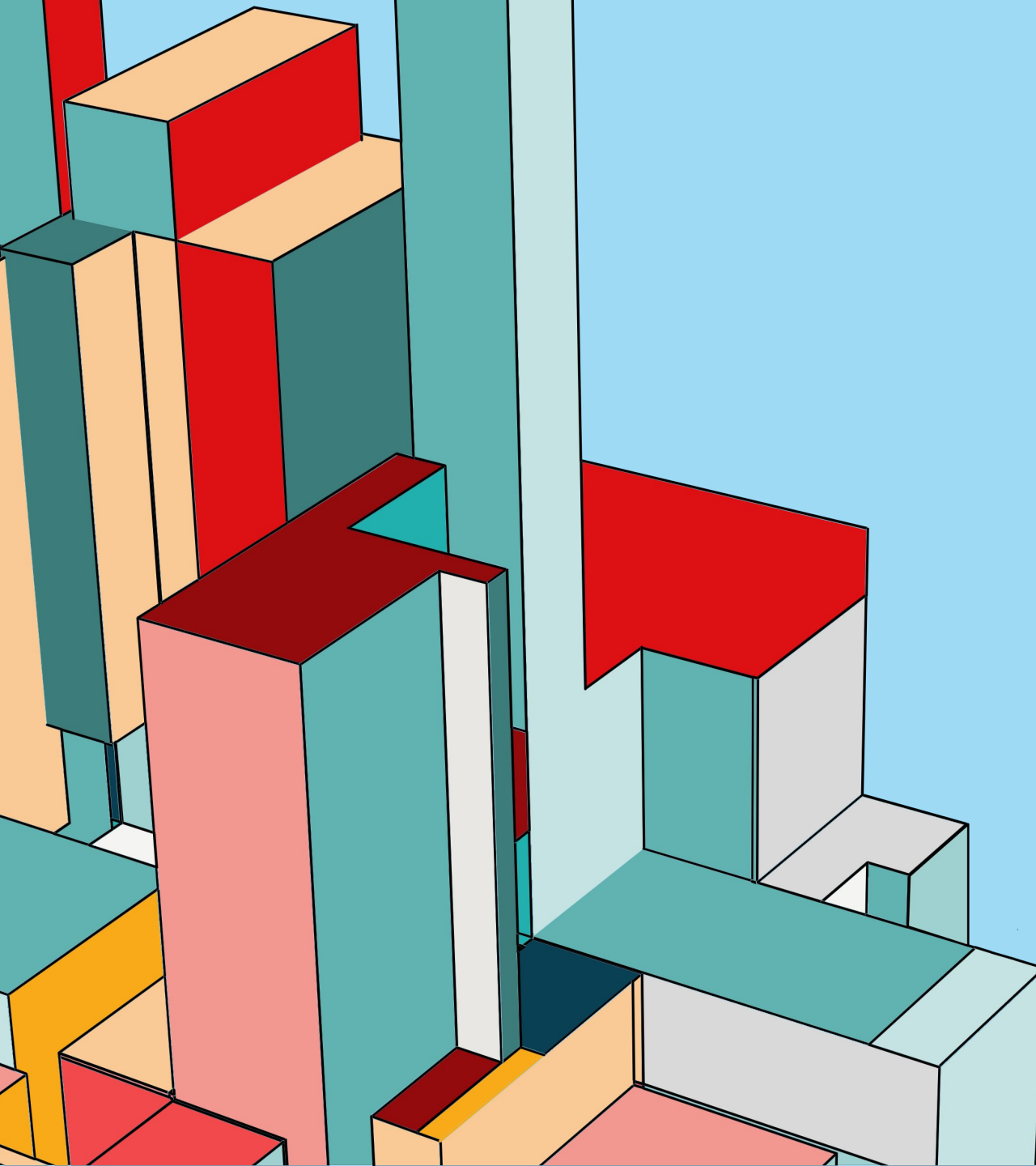
Your class timetable

← Today → Select date

Synchronise with calendar

WEEK 37	MON 7 SEP	TUE 8 SEP	WED 9 SEP	THU 10 SEP	FRI 11 SEP
08:00					08:00 Matematisk analyse 1 @ Building 1531, Aud. - room 113
09:00					08:00 Bachelor's project @ Building 1441, Aud. 3 - room 113
10:00			10:00 - 12:00 Indledende algebra 1 @ Building 1532, Undervisning Koll. G4 - room 222 Holdundervisning	10:00 - 12:00 Indledende algebra 1 @ Building 1534, Aud. F - room 125 Forelæsning	Vejledning Class
11:00					10:00 Matematik analyse 1 @ Building 1534, Aud. F - room 125
12:00		12:00 - 14:00 ARTS ad-hoc E26 @ Building 1481, Undervisning - room 341 Praktisk	12:00 - 14:00 Matematik analyse 1 @ Building 1532, Undervisning Koll. G - room 214 Holdundervisning		
13:00					
14:00	14:00 - 17:00 Matematisk analyse 1 @ Building 1536, Mat. Lab. - room 114 Praktisk	14:00 - 17:00 Indledende algebra 1 @ Building 1536, Mat. Lab. - room 114 Praktisk	14:00 - 16:00 Indledende algebra 1 @ Building 1534, Aud. F - room 125 Forelæsning	14:00 - 16:00 Matematik analyse 1 @ Building 1533, Aud. E - room 103 Forelæsning	
15:00					
16:00					
17:00					





EXERCISES FOR TODAY

EXERCISES FOR TODAY

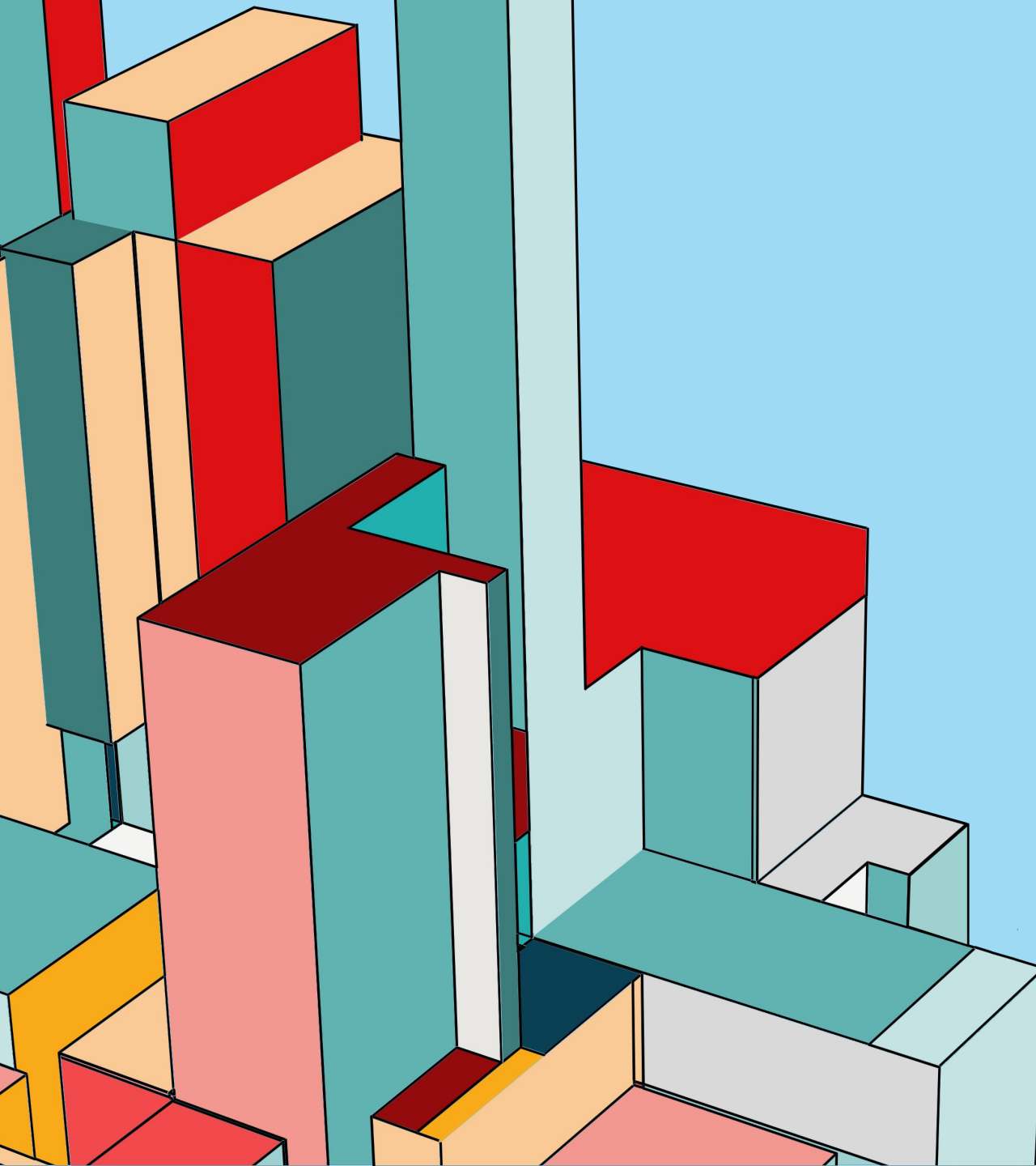
Top priority is:

- preamble_test.pdf

If you are really fast:

- OLS in hand!
- Going over Gelman and Hill (2006) chs. 1 & 2





MIDWAY DISCUSSION

OLS

$$y = X\beta + \epsilon$$

$$y = \begin{bmatrix} 1 & Winter_1 & Spring_1 & Summer_1 & Autumn_1 & temp_1 & cnt_1 \\ 1 & Winter_2 & Spring_2 & Summer_2 & Autumn_2 & temp_2 & cnt_2 \\ 1 & Winter_3 & Spring_2 & Summer_3 & Autumn_3 & temp_3 & cnt_3 \\ \dots & \dots & \dots & \dots & \dots & \dots & \dots \\ 1 & Winter_n & Spring_n & Summer_n & Autumn_n & temp_n & cnt_n \end{bmatrix} \times \begin{bmatrix} b_0 \text{ (intercept)} \\ b_1 \text{ (winter coef)} \\ b_2 \text{ (spring coef)} \\ b_3 \text{ (summer coef)} \\ b_4 \text{ (autumn coef)} \\ b_5 \text{ (temp coef)} \\ b_6 \text{ (cent coef)} \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \dots \\ \epsilon_n \end{bmatrix}$$

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$



$$\hat{\beta} = (X^T X)^{-1} X^T y$$

$$z = X^T X$$

$$z^{-1} z = I$$

$$n \cdot \frac{1}{n} = 1$$

$$\begin{matrix} 3,6 & 6,3 \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \end{matrix}^{-1} @ \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} @ \begin{bmatrix} 10 \\ 4 \\ 15 \\ -2 \end{bmatrix}$$

$$\begin{matrix} 3,3 \\ \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} \end{matrix}^{-1} @ \begin{bmatrix} 5 \\ 14 \\ 13 \end{bmatrix}$$

$$\begin{matrix} 3,1 \\ \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix} \end{matrix} @ \begin{bmatrix} 5 \\ 14 \\ 13 \end{bmatrix} = \begin{bmatrix} 2,5 \\ 7 \\ 6,5 \end{bmatrix}$$

Doing an ANOVA with OLS

Do this: $\hat{\beta} = (X^T X)^{-1} X^T y$

for a factor with 3 levels and 6 samples (so X is 6x3 with dummy coding).

See [chapter 4.6 Matrix Inversion](#) for help

$$X = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix} \quad X^T = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \quad y = \begin{bmatrix} 3 \\ 2 \\ 10 \\ 4 \\ 15 \\ -2 \end{bmatrix}$$

And if we have time we can try out Gauss-Jordan elimination

$$A^{-1}A = I \quad \text{and} \quad 2 \cdot \frac{1}{2} = 1 \quad (X^T X)^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$$

Gelman & Hill (2006) chs. 1 & 2

Models for regression coefficients

To give a preview of our notation, we write the regression equations for two multilevel models. To keep notation simple, we assume just one student-level predictor x (for example, a pre-test score) and one school-level predictor u (for example, average parents' incomes).

Varying-intercept model. First we write the model in which the regressions have the same slope in each of the schools, and only the intercepts vary. We use the notation i for individual students and $j[i]$ for the school j containing student i .¹

$$\begin{aligned} y_i &= \alpha_{j[i]} + \beta x_i + \epsilon_i, \text{ for students } i = 1, \dots, n \\ \alpha_j &= a + bu_j + \eta_j, \text{ for schools } j = 1, \dots, J. \end{aligned} \quad (1.1)$$

Here, x_i and u_j represent predictors at the student and school levels, respectively, and ϵ_i and η_j are independent error terms at each of the two levels. The model can be written in several other equivalent ways, as we discuss in [Section 12.5](#).

The number of “data points” J (here, schools) in the higher-level regression is typically much less than n , the sample size of the lower-level model (for students in this example).

Varying-intercept, varying-slope model. More complicated is the model where intercepts and slopes both can vary by school:

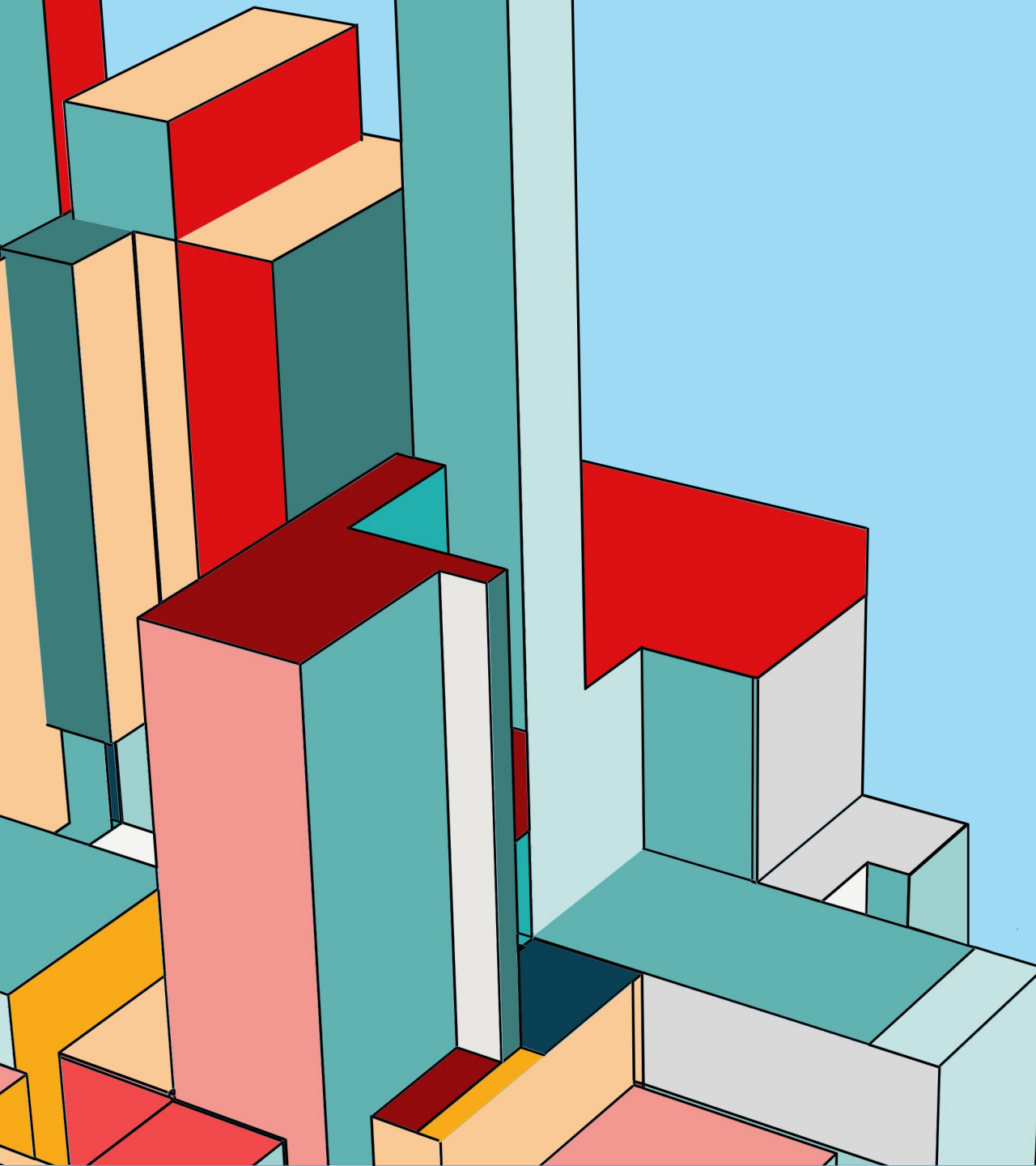
$$\begin{aligned} y_i &= \alpha_{j[i]} + \beta_{j[i]} x_i + \epsilon_i, \text{ for students } i = 1, \dots, n \\ \alpha_j &= a_0 + b_0 u_j + \eta_{j1}, \text{ for schools } j = 1, \dots, J \\ \beta_j &= a_1 + b_1 u_j + \eta_{j2}, \text{ for schools } j = 1, \dots, J. \end{aligned}$$

Compared to model (1.1), this has twice as many vectors of varying coefficients (α, β), twice as many vectors of second-level coefficients (a, b), and potentially correlated second-level errors η_1, η_2 . We will be able to handle these complications.

Some very important points on this page!

Chapter 2 reiterates some fundamental statistical terms, which I found were nice to refresh in memory :-)

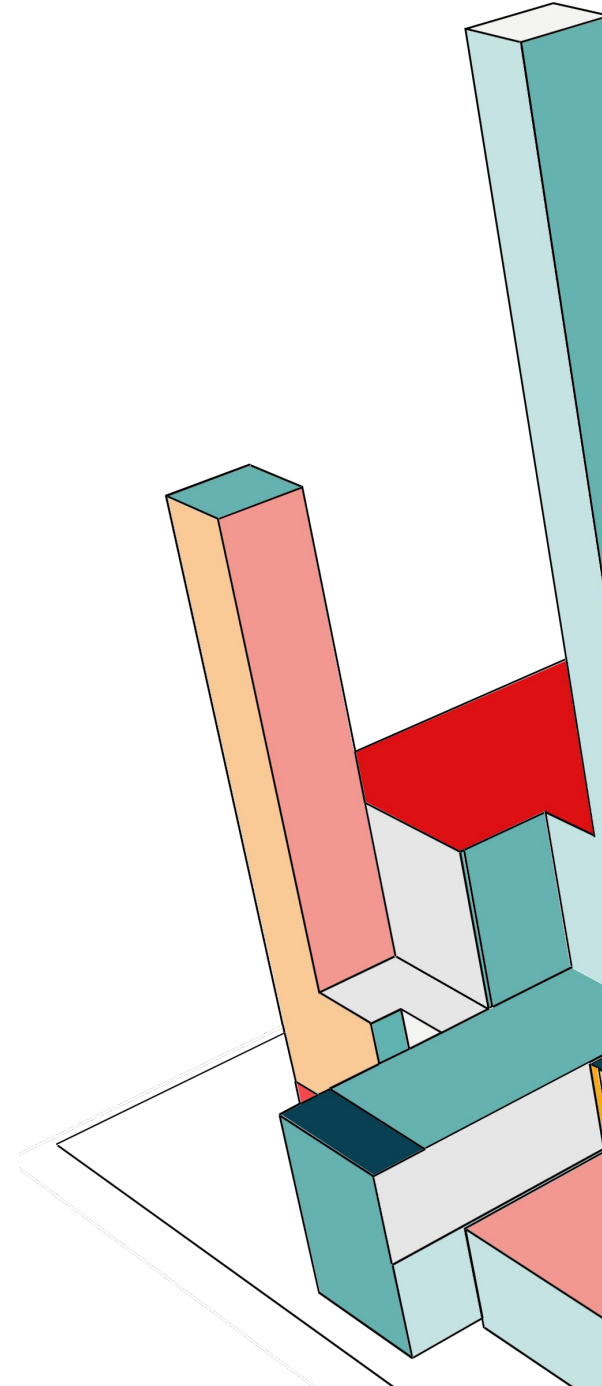




FOR NEXT WEEK

FOR NEXT WEEK

We'll catch up on Gelman and Hill (2006) chs. 1 & 2 next week :-)



SEE YOU NEXT WEEK!
insert silly picture

